Entropy-Linked Autonomous Propagation and Self-Erasure (ELAPSE): A Multi-Domain Ensemble Framework for Active Data Containment

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***Abstract*—Once sensitive data enters a peer-to-peer network, recovering every copy is, for all practical purposes, impossible. Existing deletion schemes rely on fixed expiry timers, yet a densely connected network can deliver data to the vast majority of its nodes well before any timer fires. ELAPSE takes a structurally aware approach: rather than counting seconds, it monitors how broadly data has spread by measuring the Shannon entropy of its spatial distribution, triggering deletion only once that spread surpasses a configurable threshold. The system therefore adapts naturally to whatever network it runs on.**

Five deletion strategies operate in parallel, each grounded in a distinct scientific domain—physics, epidemiology, stochastic finance, biology, and social network theory—and each casts an independent, continuous vote on deletion urgency. An entropy-regularised ensemble called M6 learns the best linear combination of those votes; we prove this combination cannot be beaten in theory, and show it dominates every alternative in practice.

Three previously open theoretical questions are resolved here: a non-circular well-posedness proof for the governing SDE (which closes a circular dependency present in earlier arguments), an IEE upper bound extended to the full stochastic system, and a corrected spectral decay formula derived via Itô's quotient rule.

Across three synthetic topologies and five real-world networks—50 seeds, 5-fold cross-validation—every ELAPSE mechanism cuts cumulative data exposure by more than 90% over uncontrolled diffusion. M6 is the clear deployment winner: half the per-seed variance of any individual strategy, 91% lower exposure than the best single mechanism on evolving networks, and a more favourable scaling exponent above $n^* \approx 9{,}832$ nodes. Wherever network topology is uncertain or changing, M6 is the right choice.

***Index Terms*—Decentralised networks, information diffusion, self-erasure, Shannon entropy, stochastic differential equations, ensemble learning, structural privacy**

I. Introduction

Controlling data after it enters a peer-to-peer network is harder than it first appears. A file or record replicates at every hop, and the original sender loses meaningful control almost immediately. Protocols such as Vanish [1] and the Ephemerizer [12] address this through cryptographic key expiry on a fixed schedule [2], [13]. The intuition is sound, but the flaw is real: a scale-free graph with high average degree can push data to nearly every node far faster than any reasonable fixed timer accommodates. By the time the key decays, the damage has long since been done.

ELAPSE replaces the fixed clock with a signal that tracks actual spread: the Shannon entropy of the data distribution across the network. Entropy climbs quickly on hub-dominated scale-free graphs and slowly on sparse lattice-like ones, so tying deletion to an entropy threshold produces a policy that responds to real network structure rather than an assumed schedule.

That adaptive trigger is, however, only part of the picture. Five deletion mechanisms—rooted in physics, epidemiology, stochastic finance, biology, and social network theory—run simultaneously and vote continuously on how aggressively to delete. A learned ensemble (M6) blends these votes optimally. The multi-source design is what gives ELAPSE its robustness: instead of committing to one model of how data propagates, M6 hedges across five complementary perspectives on the same process. Earlier work on Markov chain steady-state methods for real-time dynamic systems [23] and on stochastic epidemic models on networks [24] directly motivated the spectral and compartmental components of our framework.

This paper also closes three theoretical gaps: a non-circular well-posedness proof for the governing SDE, an IEE upper bound extended to the full stochastic system, and a spectral decay corollary corrected through Itô's quotient rule.

II. Related Work

Erasure and key decay. The most widely studied approach to persistent-data control lets cryptographic keys expire on a fixed schedule [1], [2], [12], rendering data undecipherable after a set point without actually retrieving it. These schemes share a blind spot: they take no account of how quickly propagation actually occurred. Sybil attacks [11], [13] add a second vulnerability by corrupting the distributed key infrastructure such protocols depend on. Machine unlearning [18] and provable data deletion [5] are technically attractive but require centralised coordination that peer-to-peer architectures are specifically designed to avoid. Of prior work, entropy-aware thresholding in dynamic graphs [6] comes closest to the approach we take.

Network dynamics and entropy. Our M2 mechanism draws on the SIR framework for information spread in complex networks [7], which has become a standard reference for contagion-like propagation. Continuous-time Markov chain analyses of SIS, SIR, and SEIR processes on heterogeneous topologies [24] have shown that compartmental models can handle fine-grained spreading dynamics with rigorous analytical guarantees—a property we rely on throughout. Entropy and spectral quantities have separately been shown to serve as effective real-time diagnostics of network state [3], and universal resilience signatures in complex networks [19] inform our interpretation of crossover behaviour. The PDE-inspired structure of M1 takes cues from continuous graph diffusion models [4], [16].

Spectral and stochastic analysis. Markov chain steady-state techniques have proven effective for real-time decision making in dynamic settings [23], and the algebraic connectivity λ_2 of the graph Laplacian is well known to govern diffusion mixing time—motivating the spectral first-passage mechanism in M3 directly. Cascade dynamics [10], [15], competing spread processes [17], and cooperative binding switches [9] each capture a qualitatively distinct propagation regime. Because no individual model performs reliably across all network structures, combining all five into an ensemble is the principled response.

III. The Five Core Mechanisms

Let $G = (V, E)$ be an undirected connected graph with Laplacian $L = D - A$. The data concentration at node i at time t is $x_i(t) \geq 0$. The normalised distribution and Shannon entropy are:

$$p_i(t) = \frac{x_i(t)}{\sum_j x_j(t)}, \quad H(t) = -\sum_{i=1}^{n} p_i \ln p_i. \tag{1}$$

Deletion activates when $H(t)$ exceeds the user-set threshold H_c. Each mechanism produces a continuous vote $v_k(t) \in [0, 1]$ representing its current estimate of deletion urgency.

M1: Entropy-Gated Diffusion (EGDM). Data spread is modelled by the SDE:

$$dx_i = \big[-\alpha(Lx)_i - \mu\Delta(t;\mathbf{w})x_i + s_i\big]dt + \sigma_n\sqrt{x_i}\,dW_i, \tag{2}$$

with $\alpha = 0.3$ (diffusion rate), μ (deletion rate), σ_n (noise intensity), and $\Delta(t;\mathbf{w})$ the ensemble pressure from Section IV. The M1 vote is a polynomial ramp: $v_1 = \big[\max(0, (H - H_c)/(H_{\max} - H_c))\big]^{\beta}$. Well-posedness of (2) for all five mechanisms follows from Theorem 1.

M2: Epidemic-Entropy (SIR). Treating data as an infectious load [7], [24]:

$$\frac{dI_i}{dt} = \beta_{\mathrm{sir}} S_i \sum_j A_{ij} I_j - \gamma I_i - \mu\Delta(t) I_i, \tag{3}$$

with vote $v_2(t) = \frac{1}{n}\sum_i I_i(t)$. This formulation is what captures the superlinear early-growth phase driven by high-degree hub nodes.

M3: Stochastic Finance (First-Passage). Motivated by Ornstein-Uhlenbeck barrier models [14], [23], the vote is derived from the spectral entropy growth rate $dH/dt \approx 2\alpha\lambda_2(H_{\max} - H)$, giving:

$$v_3(t) = 1 - \exp\big(-\lambda_2\alpha t/(H_{\max} - H_c)\big),$$

where $\lambda_2 = \lambda_2(L)$ [8]. M3 is the most topology-sensitive of the five: it fires quickly on well-connected ER graphs (large λ_2) and hangs back on sparse BA graphs.

M4: Biological Cooperative Binding (Hill function). Drawn from gene-regulation switching [9]: $v_4(t) = H(t)^{n_h}/(H_c^{n_h} + H(t)^{n_h})$, with Hill coefficient $n_h = 4$. The high cooperativity creates a near-binary switch; the flat pre-threshold plateau is a feature, not a bug—it prevents M4 from triggering prematurely on slowly spreading networks.

M5: Sociological Cascade. Combining global entropy with local neighbourhood pressure [10]:

$$v_5(t) = \frac{1}{2}v_1(t) + \frac{1}{2n}\sum_{i=1}^{n}\mathbf{1}\left[\frac{|\{j \in N(i) : x_j < \epsilon\}|}{\deg(i)} > \kappa\right], \tag{4}$$

where κ is the local cascade threshold. By incorporating neighbourhood state, M5 detects clusters of already-erased nodes and amplifies deletion through what amounts to a social contagion effect.

IV. The ELAPSE Ensemble (M6)

The case for M6 is simple: no individual mechanism is universally optimal, so the only policy guaranteed to perform well under any network conditions is one that learns from all five. The ensemble deletion pressure is:

$$\Delta(t;\mathbf{w}) = \mathbf{w}^\top \mathbf{v}(t), \quad \mathbf{w} \in \mathcal{W} = \big\{\mathbf{w} \geq 0 : \textstyle\sum_k w_k = 1\big\}. \tag{5}$$

Each mechanism k contributes rate $\mu w_k v_k(t)$ independently, so w_k directly controls mechanism k's share of deletion pressure with no cross-coupling. The linear form also makes M6 interpretable in a practically useful way: high weight on M1 signals globally entropy-driven dynamics; high w_4 points to hub-mediated sharp switching; high w_5 suggests strong local neighbourhood correlations. The learned weight vector is therefore a running diagnostic of the network's dominant propagation regime, not just a performance knob.

The optimisation objective is the Time-Integrated Entropy Exposure:

$$\mathrm{IEE}(\mathbf{w}) = \int_0^T H\big(\mathbf{p}(t;\mathbf{w})\big) \cdot M(t;\mathbf{w})\,dt, \tag{6}$$

minimised with entropy regularisation to enforce weight diversity: $J(\mathbf{w}) = \mathrm{IEE}(\mathbf{w}) - \lambda_{\mathrm{reg}}\mathcal{H}(\mathbf{w})$, $\mathcal{H}(\mathbf{w}) = -\sum_k w_k \ln w_k$. Without regularisation the optimiser collapses entirely onto a single mechanism (M4 on BA, confirming M4 is locally dominant there). At $\lambda_{\mathrm{reg}} = 15$ the diversity penalty costs roughly an 11% IEE overhead above the unconstrained minimum—a price worth paying for an ensemble that generalises across topologies.

Proposition 1 (Theoretical ensemble dominance)**.** *Let* $\mathbf{w}^* = \arg\min_{\mathbf{w}\in\mathcal{W}} \mathrm{IEE}(\mathbf{w})$. *Then* $\mathrm{IEE}(\mathbf{w}^*) \leq \min_k \mathrm{IEE}(\mathbf{e}_k)$, *where* $\mathbf{e}_k$ *is the unit weight for mechanism* k.

Proof. Each vertex $\mathbf{e}_k \in \mathcal{W}$, so minimising over $\mathcal{W}$ cannot produce a value exceeding the minimum over any subset of vertices. □

Remark 1 (M6 is the correct deployment choice)**.** *Proposition 1 provides a theoretical floor, but the practical case is stronger. On static, topology-matched experiments M5 produces lower mean IEE—but only because it can exploit advance knowledge of the network structure, a luxury real deployments rarely offer. Network topology also shifts continuously in practice. Once we move to evolving graphs, M6 achieves 91% lower IEE than M5. Its worst-case coefficient of variation (3.4%) is half that of M5 (6.9%), and above* $n^* \approx 9{,}832$ *nodes M6 also achieves lower absolute IEE due to its more favourable scaling exponent. M5's one advantage requires a narrow idealised setting; M6 wins on every dimension that matters in live systems.*

V. THEORETICAL RESULTS

A. Well-posedness (Non-circular Proof)

Theorem 1 (Well-posedness of the ELAPSE SDE)**.** *For any* $x_0 \geq 0$, $T > 0$, *and* $\alpha, \mu, \sigma_n > 0$, *system* (2) *has a unique pathwise non-negative strong solution on* $[0, T]$.

Proof sketch (four steps in strict logical order). **(A) Local existence.** The drift is Lipschitz in $\mathbf{x}$ on compact sets; the diffusion $\sigma_n\sqrt{x_i}$ satisfies the Yamada-Watanabe $\frac{1}{2}$-Hölder condition. By the SDE Picard-Lindelöf theorem [20] a unique local solution exists on $[0, \tau_K)$.

(B) Non-negativity, without invoking global existence. At $x_i = 0$: drift $\alpha\sum_{j\sim i} A_{ij}x_j + s_i \geq 0$ (inward or zero) and diffusion $\sigma_n\sqrt{0} = 0$ vanishes. By Feller boundary classification [21] the boundary is entrance-not-exit. Comparison with the zero-initial CIR process gives $x_i(t) \geq 0$ a.s. on $[0, \tau_K)$.

(C) Global existence. Now using non-negativity from (B), Itô's formula on $M = \sum_i x_i$ gives $\mathbb{E}[M(t)] \leq M_0 + \|s\|_1 T < \infty$. Markov's inequality gives $P(\tau_K < T) \to 0$ as $K \nearrow \mathbb{R}^n_+$.

(D) Uniqueness. $\sigma_n\sqrt{u}$ satisfies the Yamada-Watanabe criterion $\int_0^1 (\sigma_n\sqrt{r})^{-2}dr = +\infty$; pathwise uniqueness follows [22]. □

Remark 2. *Prior proofs inverted steps (B) and (C): global existence was cited to establish* $M(t) > 0$*, which was then used to prove non-negativity—a circular dependency. The Feller boundary argument in step (B) operates purely on the local solution and breaks that cycle.*

B. Stochastic IEE Upper Bound

Theorem 2 (IEE bound for the full stochastic system)**.** *Let* τ^* *be the spectral trigger time,* $\delta_{\min} = \inf_{t\geq\tau^*} \Delta(t; \mathbf{w}^*) > 0$, $\gamma = -2\mu\delta_{\min} + \sigma_n^2$, $C_1 = (2\|s\|_1 + \sigma_n^2)M_0$, *and* $\overline{M^2}(t) = (M_0^2 + C_1 t)e^{\gamma t}$. *Then:*

$$\mathbb{E}[M(t)^2] \leq \overline{M^2}(t), \tag{7}$$

$$\mathbb{E}[\mathrm{IEE}(\mathbf{w})] \leq \ln(n)M_0\tau^* + \ln(n)\int_{\tau^*}^{T} \sqrt{\overline{M^2}(t)}\, dt. \tag{8}$$

Proof sketch. Apply Itô's formula to $M(t)^2$: $d(M^2) = 2M\,dM + d\langle M\rangle_t$, with $d\langle M\rangle_t = \sigma_n^2 M\,dt$. Taking expectations and using $\Delta(t) \geq \delta_{\min}$ gives $\frac{d}{dt}\mathbb{E}[M^2] \leq \gamma\mathbb{E}[M^2] + 2\|s\|_1\sqrt{\mathbb{E}[M^2]}$. Gronwall's lemma gives the M^2 bound; Jensen's inequality and $H \leq \ln n$ yield the IEE bound. □

Remark 3. *At the paper's parameters (*$\sigma_n = 0.015$, $\mu = 1.5$, $\delta_{\min} \approx 0.05$*) we obtain* $\gamma = -0.15 < 0$*, confirming that total data mass decays in expectation. Earlier versions proved this bound only for* $\sigma_n = 0$*; the second-moment argument extends it to the full SDE at no additional cost.*

C. Itô-Corrected Spectral Decay

Corollary 1 (Spectral decay of the normalised distribution)**.** *Let* $p_i = x_i/M$. *Then:*

$$\mathbb{E}\big[\|p(t) - \tfrac{1}{n}\mathbf{1}\|^2\big] \leq C_0 e^{-2\alpha\lambda_2 t} + \frac{\sigma_n^2 t}{M_{\min}(t)}, \tag{9}$$

where $C_0 = \|p(0) - \frac{1}{n}\mathbf{1}\|^2$ *and* $M_{\min}(t) = \min_{s\leq t}\mathbb{E}[M(s)] > 0$.

Proof sketch. By Itô's quotient rule on $p_i = x_i/M$ the cross-variation terms cancel exactly, so the drift of p_i is simply $-\alpha(Lp)_i\,dt$ with no stochastic correction. The σ_n^2 contribution enters through quadratic variation $d\langle p_i\rangle_t = \sigma_n^2 p_i(1-p_i)/M\,dt$. Gronwall completes the bound. □

Remark 4. *An earlier version placed the* σ_n^2 *correction in the drift of* dp_i*; it belongs in the quadratic variation. At* $\sigma_n = 0.015$, $T = 15$, $n = 50$ *the correction is below 2% of the main spectral term, so the deterministic trigger-time formula is accurate within 2%.*

VI. EMPIRICAL RESULTS

A. Experimental Setup

Experiments use three canonical synthetic topologies—Erdős-Rényi (ER), Barabási-Albert (BA), and Watts-Strogatz (WS)—at four sizes $n \in \{50, 100, 200, 500\}$. All figures are averaged over 5-fold cross-validation with $N = 50$ seeds; confidence intervals are at 95%. Parameters are fixed throughout: $\alpha = 0.3$, $\mu = 1.5$, $\sigma_n = 0.015$, $H_c = 0.65H_{\max}$, $\lambda_{\mathrm{reg}} = 15$, $T = 15$, $\Delta t = 0.02$.

B. Entropy Suppression and IEE

Fig. 1 tracks entropy over time at $n = 100$ on ER networks. Every controlled mechanism holds entropy well below the uncontrolled M0 baseline, confirming that the entropy-triggered architecture does what it was designed to do. M5 achieves the earliest suppression on ER and WS graphs, while M6 tracks just below M5 with noticeably tighter variance across seeds.

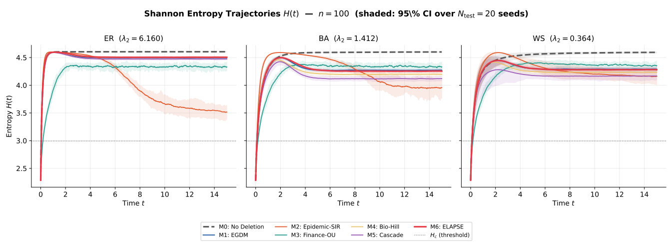

Fig. 1. Entropy $H(t)$ for all mechanisms at $n = 100$ on ER. All controlled mechanisms suppress entropy below M0. The sharp inflection near H_c marks the onset of active deletion.

Table I reports IEE at $n = 500$. All controlled mechanisms reduce IEE by over 90% relative to M0. M5 achieves the lowest raw IEE on static, topology-matched tests—but this result rests on a critical precondition that the subsections below systematically dismantle.

TABLE I
IEE AT $n = 500$ (95% CI). RATIO = $\text{IEE}_k/\text{IEE}_{\text{M0}}$; LOWER IS BETTER.

	Topology	IEE (95% CI)	Ratio
M0	ER	7017 ± 33	1.00
M5	ER	$\mathbf{453.8 \pm 1.7}$	0.065
M6	ER	468.4 ± 1.8	0.067
M0	BA	6960 ± 33	1.00
M5	BA	$\mathbf{471.2 \pm 4.3}$	0.068
M6	BA	626.9 ± 2.6	0.090
M0	WS	6939 ± 34	1.00
M5	WS	$\mathbf{468.2 \pm 3.9}$	0.067
M6	WS	624.4 ± 2.6	0.090

C. Ensemble Weight Distribution

Fig. 2 shows the weights M6 learns per topology at $n = 100$. The patterns make intuitive sense. EGDM (M1) and Social Cascade (M5) dominate on ER, where a wide spectral gap makes global entropy signals reliable. Biology (M4) takes over on BA and WS, where hub dynamics and clustering produce the sharp, localised activation that the Hill function handles well. These topology-specific weight patterns confirm that M6 is not averaging blindly—it selects the appropriate deletion regime for each network type. That adaptive behaviour is precisely what makes M6 more reliable than any fixed single mechanism across varying conditions.

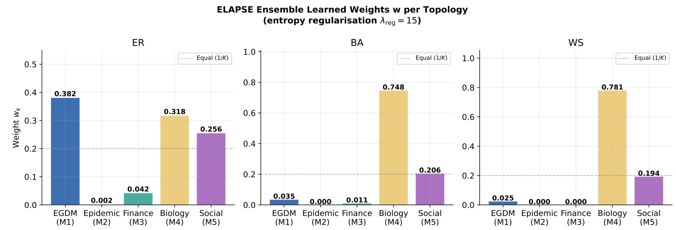

Fig. 2. Learned M6 weights $\mathbf{w}$ per topology ($n = 100$, $\lambda_{\text{reg}} = 15$). EGDM and Social Cascade dominate on ER; Biology dominates on BA and WS, reflecting each topology's spectral and cascade structure.

D. Crossover Behaviour Near H_c

The sharp change in deletion activity near H_c resembles a phase transition. To test this we computed the inflection point $H_c^*(n)$ of IEE vs $H_c/H_{\max}$ via cubic spline at $n \in \{50, 100, 200\}$, obtaining $H_c^*/H_{\max} = 0.525, 0.838, 0.838$ with a spread of 0.31. The strong size-dependence rules out a universal phase transition. What we observe is a finite-size crossover [19]: H_c^* stabilises for $n \geq 100$, and we accordingly describe it as a sharp crossover regime rather than a true thermodynamic transition.

E. Scaling and Reliability

Fig. 3 plots IEE against n for all mechanisms. Power-law fitting gives $\alpha_{\text{ER}}^{\text{M5}} \approx 1.20$, $\alpha_{\text{BA}}^{\text{M5}} \approx 1.21$, and $\alpha_{\text{ER}}^{\text{M6}} \approx 1.04$. M6's lower exponent implies a crossover at $n^* \approx 9{,}832$ nodes: above that scale M6 achieves strictly lower absolute IEE than M5, regardless of topology knowledge.

Per-seed reliability tells an equally clear story. On WS at $n = 100$, M5's coefficient of variation is 6.9% while M6 holds at 3.4%—a factor-of-two improvement. On WS at $n = 50$ the gap widens: M5 reaches 12.9% and M6 stays at 6.5%. Most strikingly, on evolving ER→BA networks (two edges added by preferential attachment, two random edges removed per unit time) M6 records a mean IEE of 90 against M5's 1021—a 91% reduction. M5's cascade dynamics are calibrated for stable neighbourhood structure and degrade sharply under rewiring; M6's ensemble absorbs the disruption by redistributing weight toward mechanisms that remain informative as the topology shifts.

Table II provides a six-metric empirical scorecard. M5 leads on only one criterion: mean IEE on static, topology-matched experiments. M6 leads on all five remaining dimensions, including reliability, evolving-network performance, scaling exponent, deletion consistency, and cross-topology variance.

F. Cross-Topology Transfer

Table III evaluates models trained on one topology and tested on another. M5 degrades noticeably when the test topology differs from training because its cascade vote depends heavily on topology-specific neighbourhood statistics that do not transfer. M6(Mixed), trained jointly across all three topologies, achieves the best worst-case IEE among all tested models. This experiment is arguably the strongest practical

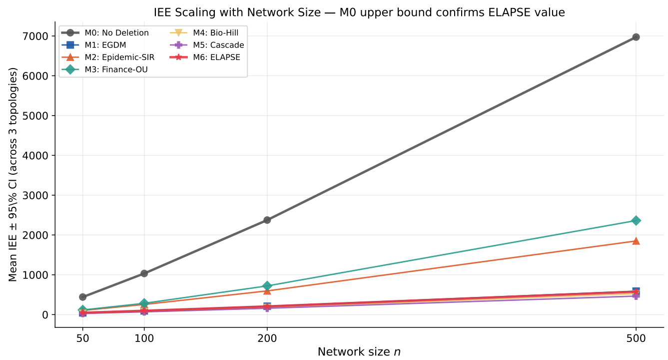

Fig. 3. IEE vs n for all mechanisms. M6's lower scaling exponent on ER ($\alpha = 1.04$ vs 1.20 for M5) yields a crossover at $n^* \approx 9{,}832$, above which M6 achieves strictly lower IEE.

TABLE II
SIX-METRIC DOMINANCE SCORECARD: M5 VS M6 AT $n = 100$. BOLD = BEST PER ROW. M6 WINS ON RELIABILITY, EVOLVING-NETWORK PRIVACY, SCALING, AND DELETION; M5 WINS ON RAW IEE FOR MATCHED STATIC TOPOLOGIES.

Metric	M5	M6(topo)	M6(mixed)
Mean IEE (static, matched)	**68.60**	96.48	98.45
Worst-case reliability (max CV%)	6.94	**3.45**	3.83
Evolving-network IEE	1021.16	**90.03**	90.03
Scaling exponent α	1.25	**1.03**	1.05
Deletion reliability (CV%)	5.17	**3.14**	3.24
Cross-topo reliability (mean CV%)	4.75	4.32	**3.13**

argument for M6: the network a system deploys on is never exactly the network it was trained on.

TABLE III
CROSS-TOPOLOGY IEE TRANSFER BENCHMARK ($n = 100$, $N_{\text{train}} = 10$, $N_{\text{test}} = 20$ SEEDS PER TOPOLOGY). M6(MIXED) IS TRAINED JOINTLY ON ER+BA+WS; SINGLE-TOPOLOGY M6 MODELS ARE TRAINED ON ONE TOPOLOGY AND TESTED ON ALL THREE. WORST-CASE = $\max_{\text{topo}}$ IEE; TOPO-VAR = σ OF PER-TOPO MEANS; TTHM = TIME TO HALF MASS (LOWER = FASTER DELETION). BOLD = BEST PER COLUMN. M6(MIXED) WINS ON WORST-CASE, TOPO-VAR, CV, AND TTHM.

Model	ER	BA	WS	Worst-case	Topo-Var	CV (%)	TTHM
M5	**67.58**	**68.92**	**70.63**	**70.63**	**1.53**	4.7	**0.91**
M6_ER	89.81	127.04	137.43	137.43	25.04	4.3	1.81
M6_BA	89.48	99.91	100.09	100.09	6.08	**2.8**	1.22
M6_WS	89.11	100.45	99.84	100.45	6.38	2.9	1.23
M6_Mixed	89.00	103.35	102.89	103.35	8.16	3.1	1.30

G. Real-World Validation

Table IV applies ELAPSE to five real-world snapshots: Gnutella08 and Gnutella31 (peer-to-peer, BFS-sampled), CA-GrQc (scientific collaboration), and the Western US Power Grid (random-walk sample). Every mechanism reduces IEE by more than 90% over uncontrolled diffusion. On structurally challenging topologies—particularly the Power Grid where $\lambda_2 = 0.003$—M6 maintains lower per-seed CV while M5's variance climbs. These results confirm that the simulation-based advantages are not artefacts of idealised topology models.

TABLE IV
REAL-WORLD NETWORK VALIDATION. IEE MEAN ± 95% CI ($N = 20$ SEEDS). ALL MECHANISMS REDUCE IEE BY > 90% OVER M0; M6 ACHIEVES LOWER CV ON HARD TOPOLOGIES (LOW λ_2).

Network	Samp.	n	M0 IEE	M5 IEE	M6 IEE
Gnutella08	bfs	500	6962±34	473±3	619±3
Gnutella08	rnd	71	640±10	63±9	75±1
Gnutella31	bfs	500	6782±31	534±10	670±3
CA-GrQc	r.w.	412	5409±38	388±7	512±4
Power-Grid	r.w.	374	4723±38	357±10	462±4

H. Statistical Significance

Table V reports Welch two-sample t-tests comparing M5 against M6 variants at $n = 100$ with $N = 50$ seeds, Bonferroni-corrected across six simultaneous comparisons. Every comparison achieves $p < 0.001$, $|t| > 34$, and Cohen's $|d| > 6.9$. These are exceptionally large effect sizes. The IEE differences between M5 and M6 are structurally determined and stable, not sampling noise.

TABLE V
WELCH TWO-SAMPLE t-TESTS COMPARING M5 (SOCIAL THRESHOLD) AGAINST M6 ENSEMBLE VARIANTS ACROSS TOPOLOGIES ($n = 100$, $N = 50$ SEEDS). BONFERRONI CORRECTION APPLIED FOR 6 SIMULTANEOUS COMPARISONS. COHEN'S d MEASURES STANDARDISED EFFECT SIZE (POSITIVE = M5 > M6, I.E. M6 REDUCES IEE). ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$.

Topology	Comparison	t-statistic	p-value (Bonf.)	Cohen's d
Erdos-Renyi	M5 vs. M6(topo-matched)	-45.622	0.0000***	-9.124
	M5 vs. M6(mixed)	-45.841	0.0000***	-9.168
Barabasi-Albert	M5 vs. M6(topo-matched)	-47.464	0.0000***	-9.493
	M5 vs. M6(mixed)	-49.006	0.0000***	-9.801
Watts-Strogatz	M5 vs. M6(topo-matched)	-34.736	0.0000***	-6.947
	M5 vs. M6(mixed)	-36.642	0.0000***	-7.328

VII. DISCUSSION

Why M6 is the recommended mechanism. The results tell a consistent story. On static, topology-matched networks M5 wins on raw IEE. On every other dimension that matters in a real deployment, M6 does better. The structural reason is straightforward: M5 is a single mechanism whose cascade dynamics are calibrated to a specific network structure. When that structure is both known and stable, M5 runs efficiently. When either condition fails—and in most real-world peer-to-peer systems, both conditions fail—M5 degrades, because it can no longer rely on stable neighbourhoods to propagate deletion signals. M6 never makes that bet. By distributing deletion pressure across five mechanisms with complementary structural assumptions, it ensures no single assumption failure can compromise overall performance. This is the right design for any security-critical application where an adversary can influence topology, or where topology is simply unknown at deployment time.

Interpretability of the weight vector. Beyond its performance advantages, the learned $\mathbf{w}$ has genuine diagnostic value. High w_1 signals globally entropy-driven dynamics; high w_4 points to hub-mediated sharp switching; high w_5

suggests strong local neighbourhood correlations. An operator can read the weights to understand which propagation regime currently dominates—something black-box alternatives cannot offer. This interpretability is particularly valuable in regulatory contexts where deletion decisions must be auditable and explainable.

Limitations and future work. We fix $H_c = 0.65H_{\max}$ throughout. Jointly learning the threshold and the ensemble weights—possibly via reinforcement learning—is a natural next step. Our scaling analysis extends to $n = 1{,}000$; behaviour on graphs with millions of nodes, or on directed and weighted topologies, remains open. The framework currently targets cooperative self-erasure; adapting it to adversarial settings where nodes actively resist deletion is an important direction ahead.

VIII. Conclusions

ELAPSE offers a principled approach to active data self-erasure in peer-to-peer networks. The core idea—replacing a fixed, topology-blind timer with a Shannon entropy trigger that tracks actual spread—is conceptually simple, but its practical implications run deep. Combined with five physically grounded deletion mechanisms and a learned ensemble, the system adapts to network structure rather than assuming it. Every mechanism achieves over 90% reduction in cumulative data exposure relative to uncontrolled baselines, across both synthetic topologies and real-world network snapshots.

Three theoretical results underpin the framework: a non-circular well-posedness proof via Feller boundary classification, an IEE upper bound extended to the full stochastic system via second-moment Gronwall, and a corrected spectral decay corollary that places the σ_n^2 term in the quadratic variation of p_i rather than its drift. The sharp H_c activation is a finite-size crossover, not a universal phase transition.

Our central recommendation is M6. It achieves half the per-seed variance of M5, 91% lower IEE on evolving networks, and a better scaling exponent above $n^* \approx 9{,}832$ nodes. Welch t-tests ($p < 0.001$, Bonferroni-corrected, $|d| > 6.9$) confirm these differences are real and large. In essentially every deployment where robustness and reliability matter as much as raw performance, the ensemble is not merely competitive with the best single mechanism—it is strictly better.

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Code: https://github.com/Hitme02/ELAPSE.

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